# Analytics Suite Explained

## `_compute_trend_series`

### What this module does (in one line)

It tells you whether the **emotion for an Experience Driver is rising, falling, or flat** over a chosen window — and how strong that move is.

### Inputs you can tweak

- `series`: the ERI values over time (e.g., daily ERI for 28 days).
- `min_n=3`: need at least 3 points to judge a trend.
- `smooth_min_n=7`: if we have 7+ points, apply gentle smoothing (EWMA) to reduce noise.
- `ewma_alpha=0.2`: smoothing strength (lower = smoother).
- `pct_thresh=5.0`: minimum % move (of the full ERI band) to call it up/down.
- `snr_thresh=0.75`: minimum signal-to-noise ratio to trust the move.
- `signal_presence_pct` & `min_presence_gate=0.05`: if coverage is ultra-low (<5%), force **Stable** (be humble).

### What it returns

A tiny dictionary:

- `symbol`: `"↑"` (rising), `"↓"` (falling), or `"→"` (stable).
- `trend_pct`: how big the move is, as **% of the full ERI band** (−100..+100 = width 200).
- `trend_snr`: how clean that move is (signal-to-noise). Higher = more trustworthy.

## How it works (step-by-step)

1. **Clean the data**
   Drops NaNs/inf; requires at least 3 points.

2. **Presence gate (optional)**
   If `signal_presence_pct < 5%`, we **don't trust** the window → return **Stable** with zeros.

3. **Fill gaps + (maybe) smooth**
   Linear interpolation for missing days; if we have ≥7 points, apply a light EWMA to reduce jitter.

4. **Quick "totally flat" check**
   If the series barely moves, label **Stable**.

5. **Fit a straight line across the window**
   - Compute the **slope** (ERI units per day).
   - Convert slope into **total change** over the whole window (`delta = slope * (n-1)`).

6. **Normalize to % of ERI band**
   - The ERI band is 200 units wide (−100..+100).
   - `trend_pct = (delta / 200) * 100` → now your move is a percentage of the entire possible ERI range.

7. **Compute SNR (signal-to-noise)**
   - Noise = day-to-day wiggle (std of diffs), scaled by √n (longer windows get stricter).
   - `trend_snr = |delta| / (noise * √n)`.

8. **Classify**
   - If `trend_pct > +5%` **and** `snr ≥ 0.75` → `"↑"`
   - If `trend_pct < -5%` **and** `snr ≥ 0.75` → `"↓"`
   - Else → `"→"` (not enough size or not clean enough).

### How to read the output (examples)

- `{"symbol":"↑","trend_pct":7.8,"trend_snr":1.1}`
  Rising trend, decent size (7.8% of the full ERI band) and clean enough to trust.

- `{"symbol":"→","trend_pct":4.9,"trend_snr":0.6}`
  The move is small and/or noisy → treat as **Stable** (don't overreact).

- `{"symbol":"↓","trend_pct":-12.3,"trend_snr":0.9}`
  Clear downward drift of meaningful size with acceptable clarity.

### Why this is strategically significant

- **Window-agnostic:** works the same for 7, 28, or 75-day views — so you can standardize dashboards.

- **Noise-aware:** won't trigger false alarms from jitter (EWMA + SNR + thresholds).

- **Business-readable:** a single symbol + two numbers that any exec can grasp.

- **Comparable across drivers:** `% of ERI band` puts all Experience Drivers on the same scale.

### When to act (decision cues)

- **↑ with decent SNR** → consider **Amplify/Optimize** moves around that driver (double-down, reinforce what's working).

- **↓ with decent SNR** → **Fix** posture (run RCA, prioritize friction removal).

- **→ or low SNR** → **Watch** posture (don't burn cycles; monitor until the move becomes real).

## `_compute_momentum_series`

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### What this module does (in one line)

It measures **how fast and in which direction emotional sentiment is moving** for an Experience Driver, by comparing the **first half vs. second half** of the time window.

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## Inputs you can tweak

- `series` – ERI values over time (daily or evenly spaced).
- `signal_presence_pct` – % of time slots in the window that actually have data.
- `min_presence_gate` – Minimum coverage to trust the reading (default from helper = 5%).

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### What it returns

A dictionary with:

- `symbol` – Momentum icon: `"↑↑"`, `"↑"`, `"→"`, `"↓"`, `"↓↓"`.
- `delta` – % change (first half → second half) as % of full ERI band.
- `label` – Human-readable classification (“Strongly Rising”, “Stable”, etc.).
- `description` – Short text explaining the momentum state.
- `snr` – Signal-to-noise ratio; higher = more trustworthy.
- `n_days` – Days in the data window.
- **Extras** – Emoji, strength score, polarity (+/–), thresholds used, and whether presence gate was triggered.

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### How it works (step-by-step)

1. **Clean the data**
   Removes NaNs/infinite values. Requires at least **4 points** (so both halves have something).

2. **Presence gate**
   If `signal_presence_pct < gate` (usually 5%), return **Stable** immediately — avoids overreacting to ultra-sparse data.

3. **Fill gaps + smooth (optional)**
   Linear interpolation for missing days; apply EWMA smoothing if ≥7 points.

4. **Split into halves**

   - **First half**: start to midpoint
   - **Second half**: midpoint to end
   - Take average ERI for each half.

5. **Compute raw change**
   `delta_raw = avg_second_half - avg_first_half` (in ERI units).

6. **Normalize to % of ERI band**
   Converts to % of full −100..+100 ERI range (200 wide).

7. **Signal-to-noise ratio (SNR)**

   - Calculates standard error for each half's mean.
   - Combines into a pooled SE.
   - `snr = |delta_raw| / pooled_se`.

8. **Classify using thresholds** (from `momentum_thresholds()`):

   - ↑↑ = large positive change & strong SNR
   - ↑ = moderate positive change & moderate SNR
   - ↓↓ = large negative change & strong SNR
   - ↓ = moderate negative change & moderate SNR
   - → = anything else (small, noisy, or unclear).

9. **Enrich with metadata** from `mom_details()` so you get emoji, polarity, and recommended action tone.

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## How to read the output (examples)

- `{"symbol":"↑↑","delta":25.4,"snr":1.6,"label":"Strongly Rising"}`
  → Big, clean upward move in emotional engagement — act fast to amplify.
- `{"symbol":"→","delta":3.1,"snr":0.5,"label":"Stable"}`
  → Change is too small or too noisy to trust — keep watching.
- `{"symbol":"↓↓","delta":-18.2,"snr":1.3,"label":"Strongly Falling"}`
  → Significant emotional deterioration — urgent investigation needed.

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### Why this is strategically god-tier

- **Crisp directional insight** – Tells you if the driver is heating up, cooling off, or holding steady.
- **Strength-sensitive** – SNR ensures only meaningful moves get flagged.
- **Action-mapped** – Each state can map directly to a strategic posture (Fix, Optimize, Amplify).
- **Fast to interpret** – A symbol, % change, and label is enough for decision-makers to act without deep analytics training.

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### When to act

- **↑↑ / ↑** – Positive momentum → lock in gains, amplify customer delight, expand winning experiences.
- **↓↓ / ↓** – Negative momentum → prioritize investigation, fix root causes, and stop emotional leakage.
- **→** – Stable → deprioritize for now, monitor for changes.

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## `_compute_volatility_series`

## What this module does (in one line)

It measures **how shaky or steady** the emotion is for an Experience Driver — i.e., **how much it wiggles around** day to day.

### Why it matters (strategic punch)

- **Stable = predictable** → scale, optimize, and commit resources confidently.
- **Fluctuating = fragile** → test, stage rollouts, monitor closely.
- **Highly fluctuating = risky** → pause big bets, fix root instability first (data, ops, comms).

### Inputs that shape behavior

- `series` — ERI values over time.
- `detrend=True` — removes any overall rise/fall so we measure **wiggle**, not slope.
- `robust=False` — if `True`, uses MAD→σ for **outlier resistance**.
- `signal_presence_pct` — how much of the window has real data (penalizes sparse series).
- `series_data_complete` — flag if the series was stitched/partial (small haircut).
- `smooth_min_n=7`, `ewma_alpha=0.2` — optional smoothing when enough points exist.

### What it returns

{

"tier": "✅ Stable" | "⚠ Fluctuating" | "🔴 Highly Fluctuating",

"score": <raw volatility % of ERI band>,

"score_adj": <length & coverage adjusted score>,

"n_days": <window length>,

"method": "std" | "mad_sigma",

"detrended": true|false

}

- `score` = spread (σ) as a **% of full ERI band** (−100..+100 = width 200).
- `score_adj` = apples-to-apples across windows: normalized by √n and adjusted for data coverage/complete-ness.
- `tier` = readable bucket for decisioning.

## How it works (step-by-step)

1. **Clean & smooth**
   Interpolate gaps; apply EWMA if enough points.
2. **Flatness guard**
   If the series barely moves → call it **Stable** quickly.
3. **Detrend (optional, default on)**
   Remove linear slope so volatility reflects **instability**, not direction.
4. **Pick dispersion method**
   - `std` (default) or
   - `mad_sigma` (robust to outliers).
5. **Normalize to % of ERI band**
   Convert σ to a % of the full −100..+100 range.
6. **Length & coverage adjustments**
   - Divide by √n → fair vs. different window sizes.
   - Penalize sparse coverage (full credit at ≥40% presence; floor at 0.2).
   - Haircut if `series_data_complete=False`.
7. **Tiering**
   - `≤ 7%` → ✅ **Stable**
   - `≤ 20%` → ⚠ **Fluctuating**
   - `> 20%` → 🔴 **Highly Fluctuating**

## How to read it (examples)

- `{"tier":"✅ Stable","score":4.1,"score_adj":3.6}`
  → Emotion is steady; great for **Optimize/Amplify** moves with low risk.
- `{"tier":"⚠ Fluctuating","score":12.8,"score_adj":15.2}`
  → Noticeable wobble; run **smaller experiments**, watch closely.
- `{"tier":"🔴 Highly Fluctuating","score":33.5,"score_adj":28.9}`
  → Instability is high; **pause scale**, investigate causes (ops incidents, seasonality, comms spikes).

### Decision cues (what to do)

- **Stable + Positive ERI** → lock in: scale programs, commit SLAs, promote advocacy.
- **Stable + Negative ERI** → problem is consistent; **Fix** with structural changes (process/product).
- **Fluctuating** → treat as **fragile**: A/B changes, stage rollouts, add guardrails.
- **Highly Fluctuating** → **diagnose volatility sources** (supply issues, policy shifts, PR noise, sampling artifacts) before investing.

### Good defaults / toggles

- Keep `detrend=True` (prevents slope from faking “volatility”).
- Flip `robust=True` when you expect spikes/outliers (launches, outages).
- Always pass `signal_presence_pct` & `series_data_complete` from provenance for fair scoring.

## How it complements Trend & Momentum

- **Trend** = direction (up/down).
- **Momentum** = recent acceleration (back-half vs front-half).
- **Volatility** = **confidence** in direction — is the emotional path smooth or shaky?

Together, they tell you: *where it’s going*, *how fast*, and *how reliable* that path is.

## `_compute_pattern_recognition`

## What this module does (in one line)

It detects **recurring emotional rhythms** in a driver's daily ERI — **Weekly / Monthly / Quarterly** — and (for weekly) identifies a likely **"pain day."**

### Why it matters (strategic punch)

- **Recurring patterns = predictable leverage or predictable risk.**
  - If negativity reliably spikes on a certain **day/week/month**, you can **staff, message, stock, or schedule** around it.
  - If positivity recurs, you can **amplify** it (promos, outreach, feature drops) at the right cadence.
- Pattern strength/confidence separates **true seasonality** from **noise**, avoiding overfitting to random bumps.

### Inputs that shape behavior

- `series` – daily ERI (time-indexed).
- `entity_data` – raw rows for this driver (used to compute weekday stats from **observed** data).
- `lags` – which patterns to test: `["weekly","monthly","quarterly"]`.
- `min_buffer_days` – require extra days beyond the lag so detection is stable.
- `sensitivity` (via `self.sensitivity`) – `"strict"`, `"balanced"` (default), `"early_warning"` alters gates.
- `epsilon_flat` – guards against fake patterns when variance is near zero.

### What it returns

{

"has_pattern": true|false,

"pattern_type": "Weekly" | "Monthly" | "Quarterly" | "None",

"pattern_strength": <ACF score at that lag>,

"pattern_confidence": "Strong" | "Moderate" | "Weak" | "Candidate" | "None",

"pain_day": "Monday"..."Sunday" | null,

"data_coverage_days": <span analyzed>,

"min_required_days": <lag minimum>,

"eri_by_day": { "Mon": value, ... } | null,

"acf": { "lag_days": L, "score": s, "sig": ci, "gate": practical_gate }

}

- **Confirmed** patterns return `has_pattern=true` with **confidence** (Strong/Moderate/Weak).
- **Candidate** means "statistically interesting but below practical gate" — worth watching, not acting.

## How it works (step-by-step)

1. **Build a continuous daily timeline** from min→max date; **linear interpolate gaps** (no forward-fill) to avoid step artifacts.
2. **Detrend & center** the series so we measure **seasonality**, not overall slope.
3. **Flatness guards** (absolute & relative): if there's essentially no variance, exit early.
4. **Coverage & lag screening**: only test lags where we have **lag + buffer** days.
5. **Sensitivity profile** sets practical gates:
   - `strict` = fewer false positives (higher gate),
   - `early_warning` = catch weak but emerging rhythms,
   - `balanced` = middle ground.
6. **ACF-based detection** at the lag (7/30/90):
   - Compute ACF; compare lag score to **statistical CI** (white-noise bound) **and** a **practical gate** (`max(CI + delta, min_gate)`).
   - **Weekly harmonic support**: also check day 14; if strong and close to day-7, **promote** confidence (true weekly often echoes at 14).
7. **If confirmed**: set `has_pattern=true`, fill strength/confidence.
   - For **Weekly**, estimate **pain_day** (lowest mean ERI) from **observed** weekday stats when coverage is decent; else try **STL** seasonal fallback to infer day.

8. **Else if above CI but below gate**: return **Candidate** with likelihood.
9. **Final flatness guard** post-ops to avoid edge artefacts.
10. **Return**: confirmed pattern > candidate > none.

## Reading the outputs (examples)

- `{"has_pattern":true,"pattern_type":"Weekly","pattern_confidence":"Strong","pain_day":"Tuesday"}`
  → Clear weekly rhythm; **Tuesday** is consistently worst — plan staffing/comms/ops accordingly.
- `{"has_pattern":false,"pattern_confidence":"Candidate","pattern_type":"Monthly"}`
  → Early hint of a monthly pulse; **watch**, don't act yet.
- `{"has_pattern":false,"pattern_type":"None"}`
  → No repeatable rhythm; treat changes as **event-driven**, not calendar-driven.

## Decision cues (what to do)

- **Weekly pattern (confirmed)**
  - **Pain day known** → load-balance operations, proactive comms, targeted offers, SLA/crew allocations.
  - **No pain day** but weekly exists → monitor weekday coverage; improve data collection to pinpoint day.
- **Monthly / Quarterly pattern (confirmed)**
  - Align **product releases, promos, service cycles**, or **inventory** with the upswings; pre-empt fixes before the downswings.
- **Candidate**
  - Mark as **watchlist**; recheck next window. Combine with **Momentum** and **Volatility** to decide if an experiment is warranted.

## Good defaults / toggles

- Keep `balanced` sensitivity for production; switch to `strict` for high-stakes decisions, or `early_warning` for discovery.

- Ensure `entity_data` has accurate dates (observed-only) — weekday means should be **from real observations**, not interpolations.
- Require **lag + buffer** days (e.g., ≥14 for weekly) for dependable calls.

## How it complements the rest

- **Trend/Momentum** say *where it's going* and *how fast*.
- **Volatility** says *how reliable that movement is*.
- **Pattern Recognition** says *when it tends to recur* — the **calendar playbook** for operations, marketing, and support.

## _compute_momentum_saturation_insight

## What this module does (in one line)

It combines **where customers are emotionally invested right now** (*Saturation from ERI*) with **how that emotion is moving** (*Momentum symbol*) to place the driver into a **25-cell decision grid** and return a **ready-to-act playbook** plus a **confidence score**.

### Why it matters (strategic punch)

- It translates analytics into **clear posture + owner + guidance** (e.g., “Cooling Off → ⚠️ Risk → CX/Product → test re-engagement journeys”).
- It’s **context-aware**: confidence scales with **data presence**, **volatility**, and **momentum SNR**, so leaders know **how hard to push**.
- It encodes **headroom** (how much emotional upside remains) so you don’t over-invest at the ceiling or under-react in low tiers.

### Inputs shaping the call

- `eri_score` → converted to **Saturation tier** (`Very High`…`Very Low`) + **headroom** via your contract.
- `momentum_symbol` → `"↑↑" | "↑" | "→" | "↓" | "↓↓"` with strength & polarity from the momentum contract.
- Optional confidence modifiers:
  - `signal_presence_pct` → coverage factor (full credit at ~40% presence).
  - `vol_norm` (0..1) → stability factor (1-volatility).
  - `momentum_snr` → SNR factor (scaled to strong SNR gate).
  - `series_data_complete` → small haircut if stitched/incomplete.

### What it returns (shape)

```
{
 "signal_classification": {
  "saturation_index": <0..1>,
  "saturation_tier": "High|Medium|…",
  "loyalty_tier": "🟢 Loyal|🔴 At-Risk|…",
  "momentum_tier": "Strongly Rising|Stable|…",
  "combined_quadrant": "<Momentum> Momentum × <Saturation> Saturation"
 },
```

"headroom_alignment": { "tier": "...", "guidance": "...", "interpretation": "...", "eri_range": "..." },
"quadrant_interpretation": {
"quadrant_label": "Cooling Off|Optimization Zone|...",
"urgency_level": "🚨 Crisis|⚠️ Risk|🌱 Opportunity|🔁 Watch|✅ Stable",
"interpretation": "<one-line strategic narrative>"
},
"tactical_insight": {
"emotional_pulse": "...",
"battle_status": "<urgency · label>",
"strategic_reality": "<trajectory + action cue>"
},
"actionable_strategy": {
"momentum_context": "...",
"saturation_context": "...",
"trajectory_story": "...",
"action_guidance": "...",
"recommended_owner": "CX|Product|Marketing|Exec...",
"headroom_guidance": "Sustain/Amplify|Optimize|Re-engage|Correct|Crisis Repair"
},
"meta": {
"matrix_key": "<Saturation|Momentum>",
"matrix_hit": true|false,
"saturation_tier_emoji": "🏆|✅|⚖️|⚠️|❌",
"momentum_tier_emoji": "↑↑🚀|↑📈|→➖|↓📉|↓↓🧨",
"quadrant_confidence": 0..1,
"borderline": true|false,
"saturation_bin": {"lower": x, "upper": y},
"momentum_contract": {"label":"...", "strength":..., "polarity": ...},
"components": {
"sat_centrality": 0..1,
"stability_factor": 0..1,
"presence_factor": 0..1,
"snr_factor": 0..1,
"gates_used": {"presence_full_credit": 0.40, "snr_strong": 1.25}
}
}
}

## How it works (step-by-step)

1. **Saturation from ERI**
   Map `eri_score` → **saturation index** (0..1) and **tier** via `sat_from_eri()`. Also

fetch **headroom** & a tier emoji.

2. **Momentum metadata**
 Read `momentum_symbol` via `mom_details()` → label, emoji, **strength** (weight), **polarity**.

3. **25-cell matrix lookup**
 Join **Saturation tier (emoji)** × **Momentum tier (emoji_full)** against your **quadrant matrix**.

    - If there's a hit, pull: **Diagnostic_Label, Urgency_Code, Recommended_Owner, Action_Guidance**, and narrative fields.

    - If no hit (should be rare), return a safe "Unknown" guidance telling the Data Team to validate.

4. **Borderline detection**
 Measure how **central** the saturation index is within its tier bins; flag **borderline** when near edges (tier swap risk).

5. **Confidence scoring (0..1)**

    - **Base** = 0.40×momentum_strength + 0.35×saturation_centrality + 0.25×stability_factor(=1−vol_norm).

    - **Modifiers** = × presence_factor (coverage vs full-credit threshold) × snr_factor (vs strong SNR gate).

    - **Haircut** if `series_data_complete=False`.
     Output: `quadrant_confidence` (rounded 0..1).

6. **Assemble payload**
 Return **classification**, **quadrant interpretation**, and a **ready-to-act strategy block** with owner & guidance, plus **meta** for auditability.

## How to read it (quick scenarios)

- `High × ↓` → **"Cooling Off" (⚠️ Risk, CX/Product)**

    - Confidence 0.72, not borderline → **Re-engage** tactics, test emotional freshness.

- `Very High × ↓↓` → **"Loyalty Collapse Risk" (🚨 Crisis, CX Leadership)**

    - Confidence 0.83 → activate **recovery program**; human-in-loop outreach.

- `Low × ↑↑` → **"Breakthrough Opportunity" (🌱 Opportunity, CX/Marketing)**

    - Confidence 0.65, borderline true → high upside; **celebrate early wins**, nurture carefully.
- `Medium × →` → **“Balanced Neutral” (🔁 Watch, CX)**
    - Confidence 0.58 → **observe**, don’t waste cycles; wait for movement.

### Decision cues (what to do)

- Use **Urgency Code** to set **SLA + escalation**.
- Use **Recommended Owner** to route to the right team.
- Use **Headroom Guidance** to pick the **stream posture** (Amplify / Optimize / Re-engage / Correct / Crisis Repair).
- Use **Borderline** + **Confidence** to manage **risk of false calls** (watch if borderline, act harder if high-confidence).

### How it fits with the rest

- **Trend**: long-window direction
- **Momentum**: near-term push/pull
- **Volatility**: reliability of movement
- **Pattern**: calendar rhythm
- **Momentum × Saturation Insight (this)**: **play-calling layer**—turns diagnostics into **who does what, how urgently, and why**, with a measurable **confidence**.

## `_compute_signal_strength` (QSSI)

### What this module does (in one line)

It turns **movement (trend + momentum)** and **headroom (saturation)** into a single **0–10 Signal Strength score** with a readable tier (No/Weak/Emerging/Strong/Critical) — plus full audit of *why*.

## Why it matters (strategic punch)

- One number tells leaders **how loud this driver's signal is right now**.
- It blends **velocity** (is emotion shifting, and how decisively?) with **headroom** (how much emotional space is left).
- It's **data-aware** (coverage, volatility, SNR), so you don't get fooled by noisy blips.

## Inputs that shape behavior

- `trend` – "↑" | "→" | "↓" from the Trend module.
- `momentum` – "↑↑" | "↑" | "→" | "↓" | "↓↓" (validated via contract).
- `saturation_index` – 0..1 (from ERI), mapped to **headroom tier** via contract.
- Optional magnitude/quality:
    - `trend_pct` (% of ERI band), `momentum_delta` (% of ERI band),
    - `n_days`, `vol_norm` (0..1; use adjusted volatility),
    - `signal_presence_pct`, `momentum_snr`, `series_data_complete`.

## What it returns (shape)

{

"velocity_component": { "trend_symbol": "↑", "momentum_symbol": "↑↑",

"velocity_score": 0..6, "velocity_rationale": "..." },

"saturation_component": { "saturation_index": 0..1, "saturation_score": 0..4,

"loyalty_tier": "Champion|Loyal|Neutral|Vulnerable|At-Risk",

"headroom_guidance": "Amplify|Optimize|Re-engage|Correct|Crisis Repair" },

"qssi_summary": {

"qssi_score": 0..10, "qssi_tier": "❌|🔁|🌱|🔥|💥",

"qssi_interpretation": "...", "qssi_strength_norm": 0..1,

"dominant_driver": "velocity|headroom"

},

"audit": { thresholds_used: {...}, factors: {...}, headroom: {...} }

}

## How it works (step-by-step)

1. **Sanitize & contracts**
   Validate symbols; clamp `saturation_index`. Load thresholds from your **single source of truth** (`momentum_thresholds`, `mom_details`, `sat_from_si`).

2. **Velocity base (0..6)**
   Using **trend × momentum** symbols only (no magnitudes), assign a base:

   - Trend present + **↓↓** → 6 (sharp counter-push)
   - Trend present + **↑↑** → 5
   - Trend present + **↑/↓** → 4
   - Trend present + **→** → 2
   - Trend flat + **↑↑/↓↓** → 3
   - Trend flat + **↑/↓** → 1
   - Else → 0
     *(Each rationale is recorded.)*

3. **Micro-nudges (±)**
   +1 if `|trend_pct| ≥ qssi_trend_bump` (default 10%).
   +1 if `|momentum_delta| ≥ qssi_mom_bump` (default 20%).
   −1 if `n_days < 10`.
   −1 if `vol_norm > 0.8`.
   Clamp to 0..6.

4. **Data-aware scaling**
   Multiply by:

   - **Decisiveness boost** = (0.7 + 0.3×momentum_strength)
   - **Presence factor** (full credit at ~40% coverage; floor 0.3)
   - **Stability factor** = (1 − vol_norm)

- **SNR factor** = momentum_snr / strong_gate (floored at 0.5)
  Small haircut if `series_data_complete=False`.
  Round to integer **velocity_score (0..6)**.

5. **Saturation (headroom) score (0..4)**
   From `sat_from_si`: maps to **loyalty tier** and **guidance**:
   - 0 = Champion (Amplify) … 4 = At-Risk (Crisis Repair).
6. **Composite QSSI (0..10)**
   `qssi = velocity_score + saturation_score` → Tier:
   - 💥 **Critical (≥9)** – extreme movement with low saturation (volatile/opportunity)
   - 🔥 **Strong (6–8)** – substantial shift; act now
   - 🌱 **Emerging (4–5)** – forming signal; track and prep
   - 🔁 **Weak (1–3)** – low strength; may self-resolve
   - ❌ **No (0)** – dormant
7. **Explainability**
   Marks **dominant_driver** = `"velocity"` vs `"headroom"`, and exposes all inputs & gates in `audit`.

## How to read it (quick scenarios)

- **QSSI 9–10 → 💥 Critical**
  - Example: `velocity=6, saturation=3 (Vulnerable)` → strong move in a fragile zone.
  - **Action:** immediate routing to **CX/Ops**; protect customers or seize recovery.
- **QSSI 6–8 → 🔥 Strong**
  - Example: `velocity=4, saturation=2 (Neutral)` → meaningful momentum with headroom.
  - **Action:** prioritize initiative generation (Fix/Optimize/Amplify per quadrant).
- **QSSI 4–5 → 🌱 Emerging**
  - Early but real. Pair with **Pattern** & **Volatility** to decide pilot vs. watch.

- **QSSI 1–3 → 🔁 Weak**
    - Keep on the radar; avoid heavy spend unless corroborated elsewhere.
- **QSSI 0 → ❌ No Signal**
    - Deprioritize; let it bake longer.

### Decision cues (what to do)

- Use **QSSI tier** to **order the backlog** across drivers.
- Use **dominant_driver** to decide whether to:
    - **Velocity-led** (big move): act on **change dynamics** (communications, triage, recovery, amplification).
    - **Headroom-led** (low/high saturation): act on **capacity** (optimize vs. crisis repair).
- Use **audit factors** to argue **confidence** in the score (presence/SNR/volatility).

### How it fits with the rest

- **Trend & Momentum** feed **velocity**.
- **Saturation** contributes **headroom**.
- **Volatility & presence & SNR** adjust **confidence**.
- **QSSI** is the **global "how loud" meter** that PEM consumes (`qssi_strength_norm`) for short-horizon forecasting.

## PEM – Predictive Emotional Modeling

### What this module does (in one line)

It **forecasts the near-term emotional trajectory** of a driver — **Likely Escalation** vs **At Risk of Decay** vs **Stable/Inconclusive** — with a **probability**, **confidence**, **horizon (days)**, and a short **why**.

## Why it matters (strategic punch)

- Converts diagnostics into a **forward-looking call** you can plan around (staffing, comms, inventory, releases).
- Uses only Layer-2/3 signals (no magic numbers): **momentum**, **headroom/saturation**, **QSSI**, **volatility**, and **calendar patterns**.
- Confidence is **data-aware** (coverage, stability, SNR proxies via QSSI & volatility), so leaders know how hard to act.

## Inputs that shape behavior

From upstream payloads:

- **`state_of_play`**: saturation tier, momentum tier, current trajectory story, action guidance, owner.
- **`volatility`** (% of ERI band; adjusted if series incomplete) → stability.
- **Pattern signals**: `has_pattern`, `pattern_type` (Weekly/Monthly/Quarterly), `pattern_confidence`, `pattern_strength`, optional `pain_day`.
- **QSSI**: numeric `qssi_score` (0..10) → **qssi_strength** (0..1).
- **Momentum**: `momentum_symbol` or taken from `state_of_play.momentum_tier`.
- **Data quality**: `signal_presence_pct`.

## What it returns (shape)

{

"trajectory_forecast": {

"pem_trajectory": "Likely Escalation|At Risk of Decay|Recurring Decay Risk|Stable / Inconclusive",

"pem_probability": 0..1,

"pem_confidence": "Low|Moderate|High",

"confidence_score": 0.2..0.95,

"risk_class": "Opportunity|Risk|Neutral",

"horizon_days": 14|30|90 (pattern-aware),

"explanation": "...", "version": "PEM.v1.7",

"rule_trigger": "rising+strong_qssi+pattern | falling+volatility | falling+pattern | model_choice",

"basis": "esc_prob|dec_prob",

"counterfactual_pointer": "what would flip the call"

},

"signal_diagnostics": {

"has_repeating_pattern": true|false, "pattern_type": "...", "pattern_confidence": "...",

"pattern_pain_day": "Tuesday"|null, "volatility_pct": x.x,

"momentum_tier": "Strongly Rising|...", "saturation_tier": "High|...", "qssi_tier": "echo only",

"headroom_contract": { tier/guidance/eri_range/... }

},

"future_risk_profile": {

"trajectory_story": "...", "action_guidance": "...", "recommended_owner": "CX|Product|..."

},

"audit": {

"feature_vector": { qssi_strength, momentum_score, headroom_scalar, stability, pat_strength, esc_prob, dec_prob, ... },

"elasticity_rating": "High|Moderate|Low",

"consistency": { momentum_direction, pattern_supports_escalation, volatility_pressure }

}

}

## How it works (step-by-step)

1. **Normalize core signals**

- Convert **volatility% → vol_norm (0..1)** and **stability = 1 − vol_norm**.
- Pull **numeric QSSI** (`qssi_score/10 → qssi_strength`).
- Map **momentum symbol** to **strength & polarity** (↑↑=+1.0, ↑=+0.6, →=0, ↓=−0.6, ↓↓=−1.0).
- Map **saturation tier** to a **headroom scalar** (0 Champion .. 4 At-Risk → /4.0).

2. **Directional raw scores**
   - **Escalation raw (esc_raw)** grows with **positive momentum**, **QSSI strength**, **headroom (more space = more upside)**, and **stability**.
   - **Decay raw (dec_raw)** grows with **negative momentum**, **QSSI strength**, **being closer to At-Risk**, and **volatility**.
3. **Pattern fusion (gentle)**
   - If a **repeating pattern** exists, *slightly* boost the direction it supports (e.g., weekly dip strengthens **decay** risk; weekly rise strengthens **escalation**).
4. **Convert to probabilities**
   - Normalize **esc_raw** and **dec_raw** into **esc_prob/dec_prob** (sum to ~1, unless capped).
5. **Quality & prudence caps**
   - Build a **cap** based on **QSSI strength**, **presence**, **stability**, and **momentum flatness**; keep strongest side under this cap to avoid over-confidence on weak data.
6. **Horizon selection**
   - **Weekly** → 14 days; **Monthly** → 30; **Quarterly** → 90; else a default horizon (e.g., 14).
7. **Rules-first overrides**
   - **Rising + strong QSSI + pattern** → **Likely Escalation** (≥0.65).
   - **Falling + (volatility high or pattern exists)** → **At Risk of Decay / Recurring Decay Risk** (≥0.65).
   - Otherwise, pick the higher of `esc_prob` vs `dec_prob`, or **Stable/Inconclusive** if neither is decisive.

8. **Confidence tier**
    - **High**: pattern present **and** volatility ≥ 5%.
    - **Moderate**: pattern present **or** volatility ≥ 5%.
    - **Low**: neither.
    - Adjust by presence; clamp to **0.20..0.95**.
9. **Elasticity rating**
    - From momentum strength and headroom; upgraded to **High** if a **strong pattern** coexists with **Moderate** base elasticity.

## How to read it (examples)

- **Likely Escalation · p=0.72 · High confidence · horizon=14**
  Rising momentum, good headroom, weekly upswing — plan **Amplify/Optimize** moves timed to the pattern.
- **Recurring Decay Risk · p=0.68 · Moderate confidence · horizon=30**
  Falling momentum, moderate/high volatility, monthly dip — **pre-empt** with fixes/comms before the next trough.
- **Stable / Inconclusive · p=0.53 · Low confidence · horizon=14**
  No dominant anchors; maintain **Watch** posture and improve data coverage.

## Decision cues (what to do)

- **Trajectory** → sets **posture** (Opportunity vs Risk vs Neutral) and **owner**.
- **Probability & Confidence** → set **intensity** (pilot vs full rollout; alerting thresholds).
- **Horizon** → sets **timing** (when to act).
- **Counterfactual pointer** → tells you **what would flip** the forecast (e.g., "If momentum turns to ↑ and volatility drops <5%").

## Safety rails baked in

- **Presence-aware** (coverage factor).
- **Volatility-aware** (caps & stability).
- **Contract-respecting** (momentum & saturation from single sources).

- **No tier cheating** (uses **numeric QSSI**, not string tiers, for math).

### How it fits with the rest

- **Trend & Momentum** → direction and acceleration.
- **Volatility** → reliability.
- **Pattern** → calendar rhythm.
- **QSSI** → signal loudness.
- **PEM** → **near-term forecast** with probability, confidence, horizon, and "what would change the call."

## 1) "Late Delivery ETA" — Cooling Loyalty, Weekly Pain-Day

**Context (ED):** Fulfillment → Delivery ETA Accuracy

**Telemetry snapshot**

- **ERI (last 28d):** +18 → +6
- **Trend:** ↓ (−9.4% of ERI band, SNR 1.0)
- **Momentum:** ↓ (−8.2%, SNR 0.9)
- **Volatility:** ⚠ **Fluctuating** (score_adj 14.8%)
- **Pattern: Weekly (Moderate)**; **pain day: Friday**
- **Saturation (from ERI): Medium** (Headroom = "Re-engage / Nudge")
- **Quadrant (Sat × Mom): Medium × ↓ → "Churn Warning"** (⚠️ Risk, Owner: CX/Analytics)
- **QSSI: 6 → 🔥 Strong Signal** (dominant driver: *velocity*)

**PEM Forecast**

- **Trajectory: Recurring Decay Risk**
- **Probability: 0.67**, **Confidence:** Moderate, **Horizon:** 14 days
- **Why:** Falling momentum + weekly dip (Fri) + meaningful headroom to lose

**Decision play (do this now)**

- **Owner: CX + Last-mile Ops**
- **Action window: Wed–Fri** next two weeks
- **Moves:**
    1. **Fix**: Hard-gate ETA promises on Fri routes (capacity-aware ETA); add "delay risk" badge at cart.
    2. **Optimize**: Friday-specific micro-slotting (shift loads to Thu/Sat).
    3. **Comms**: Friday AM SMS pre-alerts + voucher if >30min deviation.
- **Success criteria:** Reduce Friday ERI dip ≥ 6 pts; **Volatility** ≤ 10%; **Momentum** back to →/↑ within **14 days**.
- **Counterfactual pointer:** If **Momentum** flips to ↑ and **Volatility** < 8% by mid-window, pivot to **Optimize** only (remove voucher).

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## 2) “Smart Add-On Recommendations” — Breakthrough Opportunity

**Context (ED):** Recommendation Engine → Smart Add-On Suggestions

**Telemetry snapshot**

- **ERI (last 28d):** −12 → +10
- **Trend:** ↑ (+11.6%, SNR 1.2)
- **Momentum:** ↑↑ (+24.1%, SNR 1.5)
- **Volatility:** ✅ **Stable** (score_adj 5.9%)
- **Pattern: Monthly (Strong)**—engagement spike first week of month
- **Saturation: High** (Headroom = “Optimize”)
- **Quadrant: High × ↑↑ → “Optimization Zone”** (🌱 Opportunity, Owner: Marketing/Product)
- **QSSI: 7 →** 🔥 **Strong Signal** (dominant driver: *velocity*)

**PEM Forecast**

- **Trajectory: Likely Escalation**
- **Probability: 0.74**, **Confidence:** High, **Horizon: 30 days**
- **Why:** Strong positive momentum, stable signal, repeatable monthly lift

**Decision play (scale the win)**

- **Owner: Product + Growth**
- **Action window: Days 1–7 each month**
- **Moves:**
    1. **Optimize**: Expand model to include pantry-pairing & replenishment heuristics.
    2. **Amplify**: “Bundle save” nudges on PDP/Cart during week-1; A/B auto-add trial for top 10 SKUs.
    3. **CX Comms**: In-app card: “People who bought X also loved Y (save 12%)” week-1 only.

- **Success criteria:** Add-on attach-rate +2.5 pts; **Trend** stays ↑; **QSSI** ≥ 7 for two cycles.
- **Counterfactual pointer:** If **Volatility** > 15% or **Momentum** downgrades to →, pause auto-add; keep "bundle save" only.

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## 3) "Refund & Billing Clarity" — At-Risk with Sharp Deterioration

**Context (ED):** Billing & Payments → Refund Processing Clarity

**Telemetry snapshot**

- **ERI (last 21d):** −8 → −35
- **Trend:** ↓ (−16.9%, SNR 1.3)
- **Momentum:** ↓↓ (−26.7%, SNR 1.6)
- **Volatility:** 🔴 **Highly Fluctuating** (score_adj 29.4%)
- **Pattern: None (Candidate: Weekly)** — insufficient coverage for confirmation
- **Saturation: Low → Very Low** (Headroom = "Correct / Crisis Repair")
- **Quadrant: Low × ↓↓ → "Critical Stall"** (🚨 Crisis, Owner: CX Leadership)
- **QSSI: 9 → 💥 Critical Signal** (dominant driver: *velocity*)

**PEM Forecast**

- **Trajectory: At Risk of Decay**
- **Probability: 0.78**, **Confidence:** Moderate, **Horizon:** 14 days
- **Why:** Strong negative momentum + high volatility; headroom indicates collapse risk

**Decision play (contain damage fast)**

- **Owner: CX Leadership + Payments Ops + Legal**
- **Action window: Immediate (next 72 hours)**

- **Moves:**
  1. **Fix**: Publish refund-timeline banner at Checkout/Orders; show *refund status meter* (received → processing → settled).
  2. **Ops**: Prioritize stuck refunds > 7 days; nightly reconciliation with PSP; add auto-escalation queue.
  3. **Comms**: Proactive emails to impacted cohorts; goodwill credit for >10-day cases.
- **Success criteria:** ERI improvement ≥ 12 pts in 14 days; **Momentum** from ↓↓→↓/→; **Volatility** ≤ 18%; **QSSI** ≤ 6.
- **Counterfactual pointer:** If **Pattern** later confirms weekly spikes (e.g., Mondays), staff a Monday resolution squad; pre-weekend batch settlement.

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## Why This Suite Unlocks Unfair Strategic Advantage

- You're not staring at charts—you're holding a **routing engine**: *who acts, how hard, when, and why*, with measurable reversal triggers.
- The seven analytics **reinforce each other**:
  - **Trend/Momentum** tell direction & acceleration.
  - **Volatility** tells reliability of that movement.
  - **Pattern** gives the **calendar switch** to time action.
  - **Saturation** prevents **over/under-investing** by exposing headroom.
  - **Quadrant** converts state → **owner + urgency**.
  - **QSSI** ranks **how loud** the signal is across drivers.
  - **PEM** projects **what happens next**—and what must change to flip it.